VASVI JAIN

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EDUCATION

M.S., Biology, New York University | GPA: 3.87/4.00 — Expected Dec 2026

B.S., Neural Science and Computer Science (double major), Minor in Chemistry, New York University | GPA: 3.53/4.00 — May 2025

RESEARCH EXPERIENCE

Research Assistant, Tsien Lab, NYU Langone Health | PI: Dr. Richard W. Tsien — May 2023 – Present

Master's thesis: *Corticostriatal synaptic dysfunction in Timothy Syndrome (working title)*

- Investigating cortico-striatal dysfunction in a Timothy Syndrome type 2 mouse model and its response to MEK inhibition (SL327), linking $Ca_V1.2$ dysfunction to MAPK/ERK, MeCP2, and c-Fos signaling. This work is contributing toward a manuscript.
- Perform RNAscope, IHC, confocal imaging, and western blotting on cryosectioned mouse brain tissue.
- Developed a Python image-analysis pipeline to quantify MeCP2 mRNA in Drd1+ and Drd1− striatal neurons: Cellpose nuclear segmentation, a DARPP32 classifier fit to hand-annotated cells with leave-one-animal-out validation, RNAscope dot detection, and animal-level two-way ANOVA.

Undergraduate thesis: *Investigating the role of L-type calcium channel activation in β2a subunit–CaMKII interaction*

- Generated a non-phosphorylatable β2a T498A mutant by site-directed mutagenesis; showed by FRET-FLIM that it enhances Ca^{2+}-dependent, EGTA-reversible association with CaMKIIα in HEK293T cells, with a similar trend in primary rat cortical neurons.
- Detected phosphorylation of β2a when co-expressed with CaMKIIα using Phos-tag western blotting; analyzed FLIM lifetime and colocalization data. Work contributes to a second-author manuscript in preparation.

Neuroscience Research Intern, Eikon Therapeutics — May 2026 – Aug 2026

- Generated stable Parkinson's disease–model iPSC lines via lentiviral transduction; adapted and optimized the protocol for these lines.
- Validated neuronal marker expression, ran antibody titrations, qPCR, ICC, and Incucyte live-cell imaging; contributed to assay development and reproducibility testing in 384-well format for high-throughput screening.
- Presented results three times at internal team meetings and at the company intern showcase.

Graduate Rotation Student, Klann Lab, NYU | PI: Dr. Eric Klann — Sep 2025 – May 2026

- Optimized a RAPPL-based protocol in NGN2-iNeurons to increase yield of a minor ribosomal peak seen after RNase digestion, preparing material for downstream cryo-EM to determine whether it represents collided ribosomes.
- Maintained hiPSCs and differentiated NGN2-iNeurons; ran and analyzed polysome profiles; tested ribosome–tau interactions by co-IP and western blotting.
- Mentored undergraduate researchers in cell culture and biochemical techniques.

Research Assistant, OLab, NYU Langone Health | PI: Dr. Eric K. Oermann — May 2023 – Sep 2023

- Conceived and coded the global author-collaboration network analysis and figure for a published bibliometric study.

Research Assistant, Suzuki Lab and Dillon Lab, NYU | Participant recruitment, study sessions, data coding — Sep 2022 – Sep 2023

PUBLICATIONS

Peer-reviewed

Mir JM, Kurland DB, Cheung ATM, Liu A, Shlobin NA, Alber D, Rai S, **Jain V**, Rodriguez-Olaverri JC, Anderson RCE, Lau D, Kondziolka D, Oermann EK. The evolution of pediatric spine surgery: a bibliometric analysis of publications from 1902 to 2023. *Neurosurgery Practice* (2024).

Preprints

Evans HT, Kalavai SV, Wu V, Polavarapu A, Balamoti E, Liu WJ, Mosto O, Alder D, **Jain V**, Denil E, Yu A, Assenso E, Oliveria M, Moniz T, Golhan EN, Zhang X, Sheehan CJ, Klann E. Tau-induced ribosomal collisions impair memory through the activation of the integrated stress response. *bioRxiv* (2026).

Manuscripts in preparation

Khaled HG, **Jain V**, Wang X, Tsien RW. Ca_V1–CaMKII interactions support lasting CaMKII translocation to dendritic spines. *In preparation.*

PRESENTATIONS

Jain V, Khaled HG, Tsien RW. Investigating the role of L-type calcium channel activation in β2a subunit–CaMKII interaction. Poster, NYU CNS Neural Science Festival and NYU Undergraduate Research Conference. Apr 2025

AWARDS

Dean's Undergraduate Research Fund (DURF), New York University (two awards) — 2022, 2024

TECHNICAL SKILLS

Molecular biology: Site-directed mutagenesis, cloning, PCR/qPCR, transfection, lentiviral/AAV transduction, polysome profiling
Biochemistry and histology: Western blotting (incl. Phos-tag), co-immunoprecipitation, RAPPL, RNAscope, IHC, ICC, cryostat sectioning
Cell culture: hiPSCs, NGN2-iNeurons, primary rat cortical neurons, mouse astrocytes, HEK293T; stable line generation, aseptic technique
Imaging: Confocal, FRET-FLIM, Incucyte live-cell imaging, brightfield
Computational: Python (pandas, NumPy, scikit-learn, matplotlib, Cellpose), FIJI/ImageJ, GraphPad Prism, R (learning)
Interests: Photography (portfolio: vasvi-j.github.io), reading, baking and cooking, hiking